Personal Portfolio

Khusnil Arifandi Purnama

Machine Learning Engineer | Data Analyst

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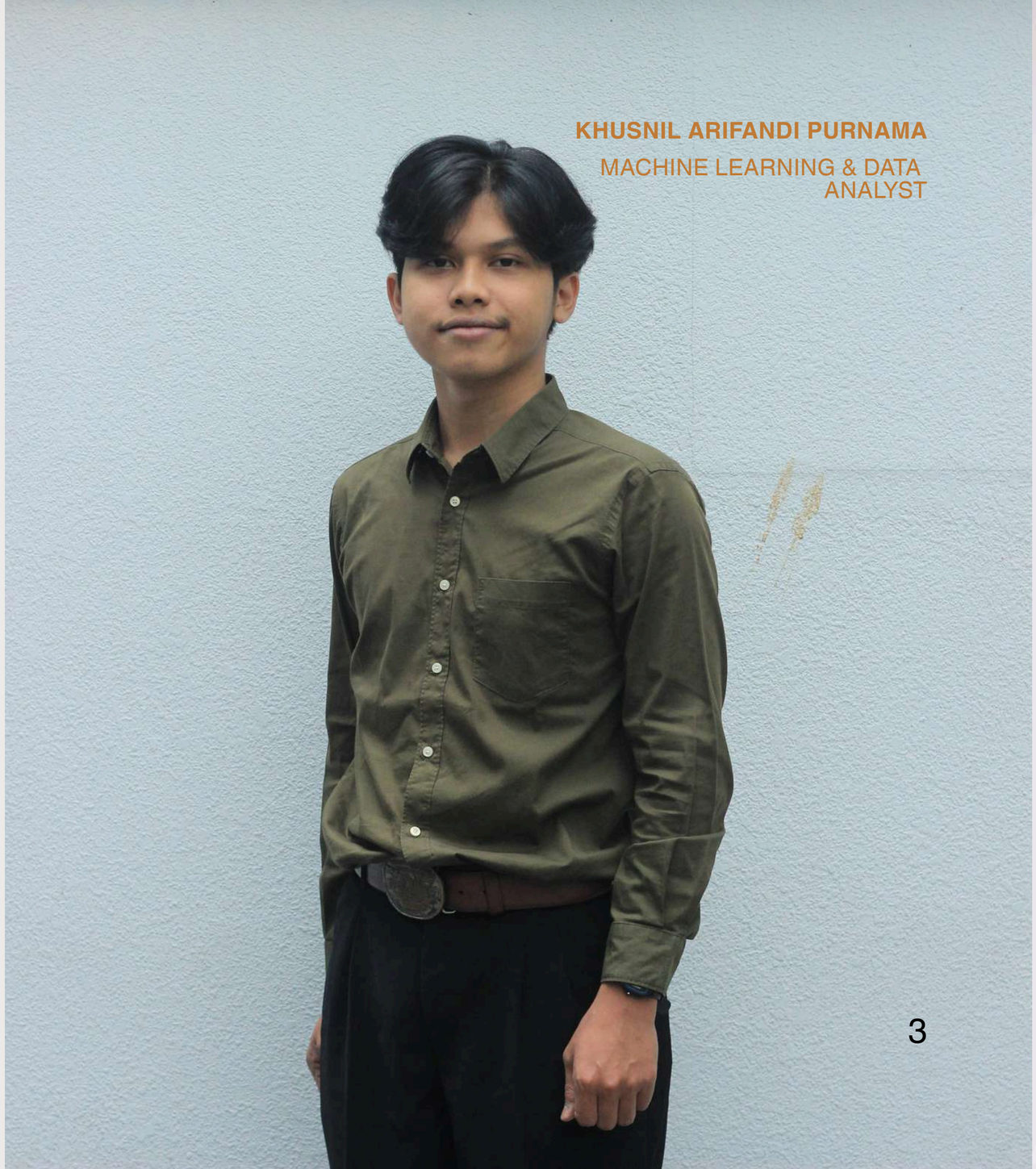
KHUSNIL ARIFANDI PURNAMA
MACHINE LEARNING & DATA ANALYST

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Who I Am

I am a data-driven Physics graduate with hands-on industrial experience crafting analytical solutions that connect complex datasets with actionable business insight

I specialize in machine learning, quantitative modelling, & statistical analysis. My approach combines analytical rigor with operational leadership – always backed by data



What I Do

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I have built data-driven solution across various domains, including industrial quality control, financial market forecasting, & operational management. Each experience has sharpened my ability to translate complex datasets & machine learning models into actionable business insights while maintaining rigorous analytical consistency

- Machine Learning
- Statistical Data Analysis
- Time-Series Forecasting
- Quantitative Modelling
- Project Management
- Business & Financial Acumen

Education

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Bandung Institute of Technology (2022-2026)
Majoring in Physics

Coding Camp Dicoding x DBS Foundation (Feb-Jul 2025)
Machine Learning Pathway
Capstone Project: SmartRecruiter

Work Experience

PT Djarum (Jul-Sep 2025)
Quality Control Internship

Organizational Experience

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Himpunan Mahasiswa Fisika (HIMAFI)
Staff of Alumni Relation & Career Development (2024/2025)
Head of Operation Intellectuelle Schule 2024

APRES! ITB
Head of Asset Management & Operation (2024-2025)

MUSTANG : After Hours by APRES! 2025
Head of Operation (Jan-May 2025)









Skills Snapshot

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Hard Skill

1. Machine Learning & NLP
2. Statistical Analysist
3. Time-Series Forecasting
4. Quantitative Modelling
5. Data Visualisation
6. Business & Market Analysist
7. Financial Statement Analysist

Soft Skill

1. Operational Leadership
2. Problem Solving
3. Project Management
4. Cross-Functional Collaboration
5. Analytical Thinking
6. Attention to Detail
7. Strategic Communication



My Process

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Problem Scoping

Defining business goals, KPIs, & target metrics

1

Data Processing

Data extraction, cleaning & EDA

2

Modelling

Algorithm selection, model training & validation

3

Business Insight

Translating data pattern into actionable market strategies

4

Deployment & Optimization

Model Implementation, A/B testing, & continuous monitoring

5

Project

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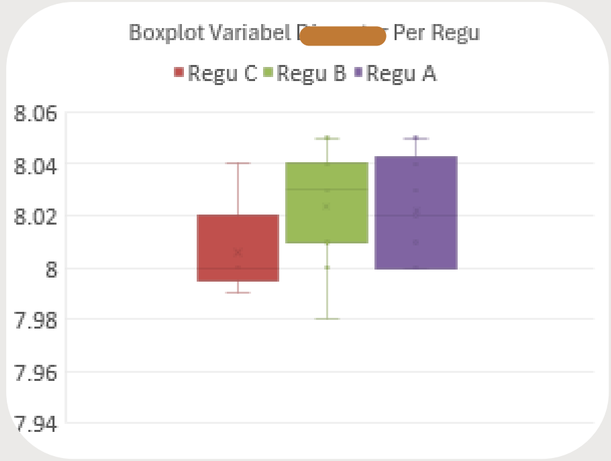
Production Stability

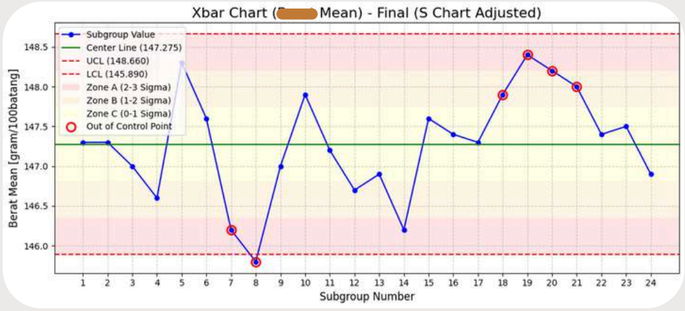
Company

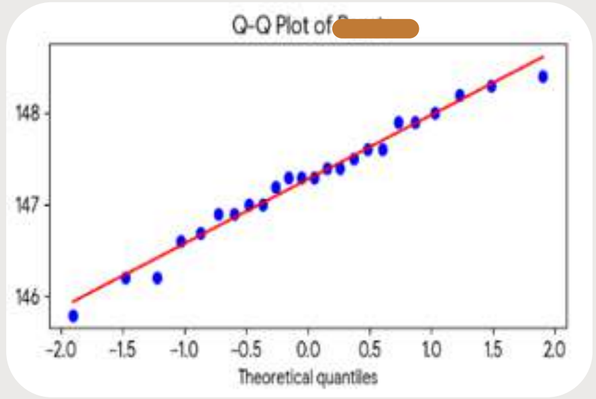
PT Djarum

Quality Control Internship

I evaluated production stability through analytical and statistical modeling using Python and Excel — executing daily quality inspections, calibrating QTM measurement machines for data accuracy, and designing a proactive defect-prevention dashboard to deliver data-driven insights directly to Factory Manager & Head of Production









Project

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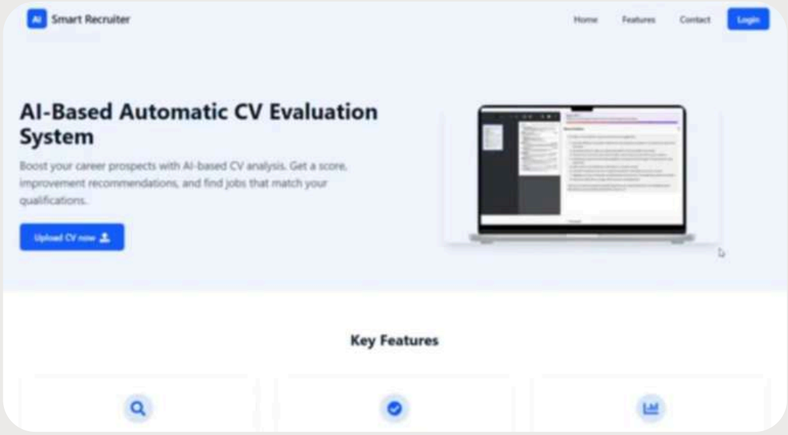
SmartRecruiter-AI CV Reviewer

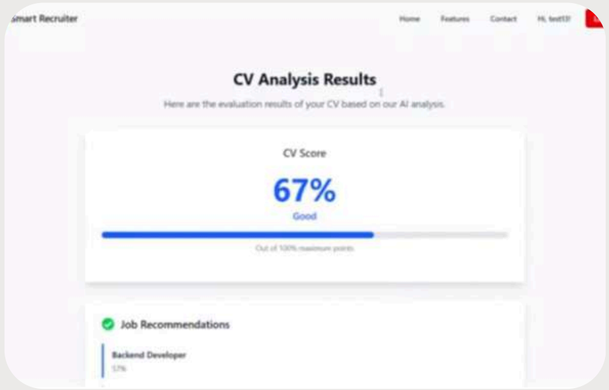
CODING CAMP powered by DBS FOUNDATION
ROAD TO FUTURE WORKFORCE

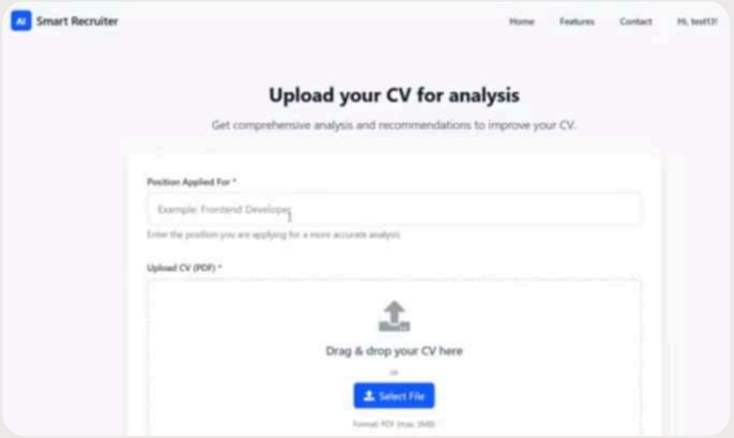
Dicoding x DBS Foundation Capstone Project

- Partnered as an ML Engineer in a 6-member cross-functional team (3 Web Developers, 3 ML Engineers) on the SmartRecruiter platform.
- Engineered Sentence-BERT (SBERT) algorithms to automate resume screening and semantic candidate scoring.
- Successfully integrated and deployed the NLP pipeline into a production-ready web application.









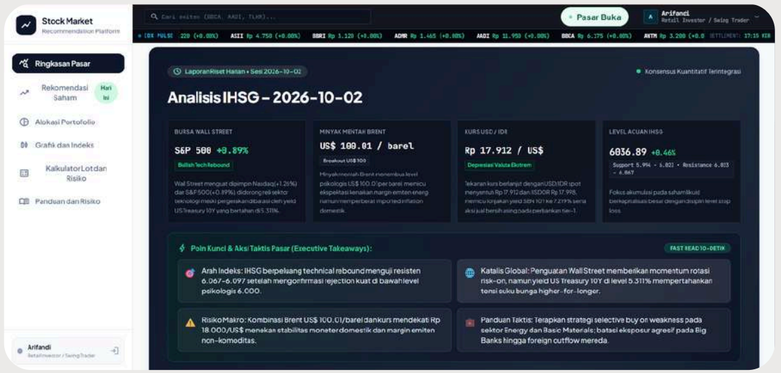
Project

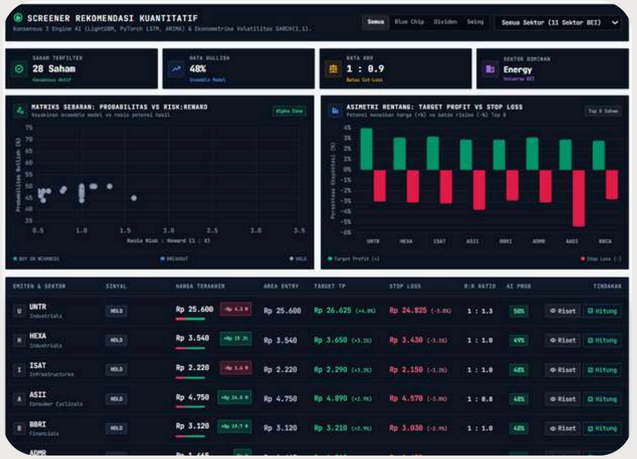
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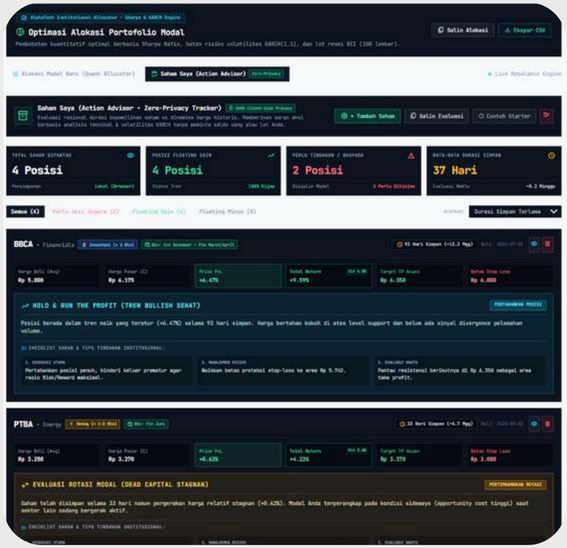
Stock Market Recomendation Website

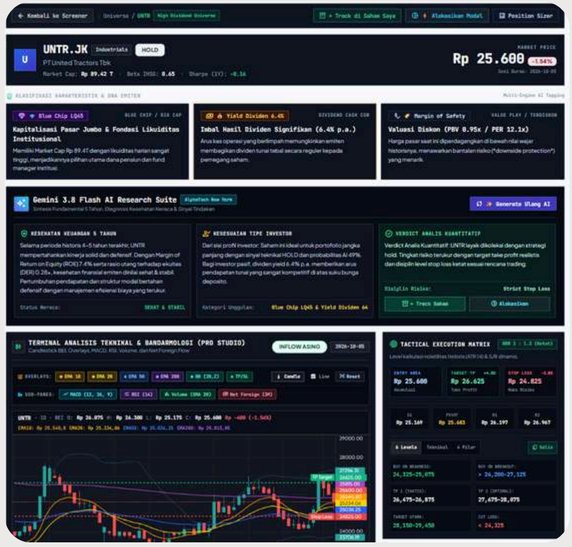
- Architected an institutional-grade equity research terminal covering 66 IDX equities.
- Built a 32-feature quantitative engine blending CatBoost, HistGB, PyTorch LSTM, and ARIMA with Student-t GARCH(1,1) risk modeling.
- Achieved 57.16% Top-Decile Precision and a +3.56% alpha gain across 17,500+ out-of-sample observations.
- Delivered real-time multi-pane TradingView analytics and dynamic ATR-14 position sizing for disciplined risk control.

https://stock-market-recommendation-silk.vercel.app







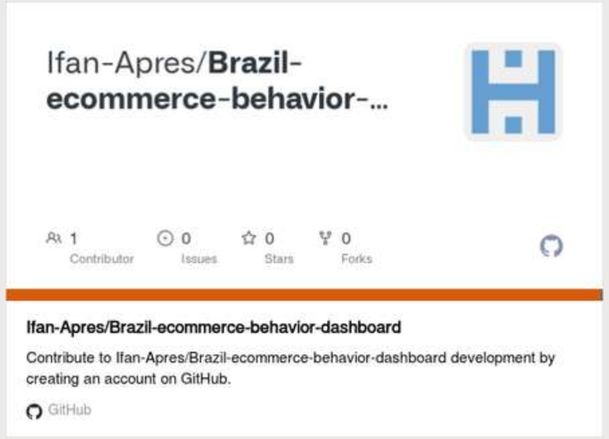


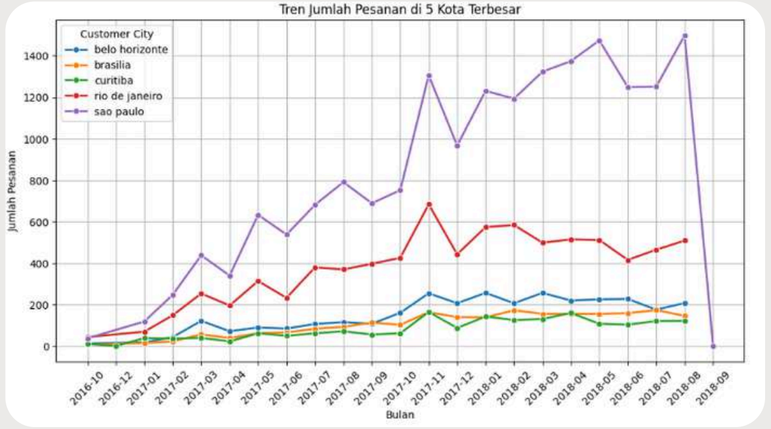
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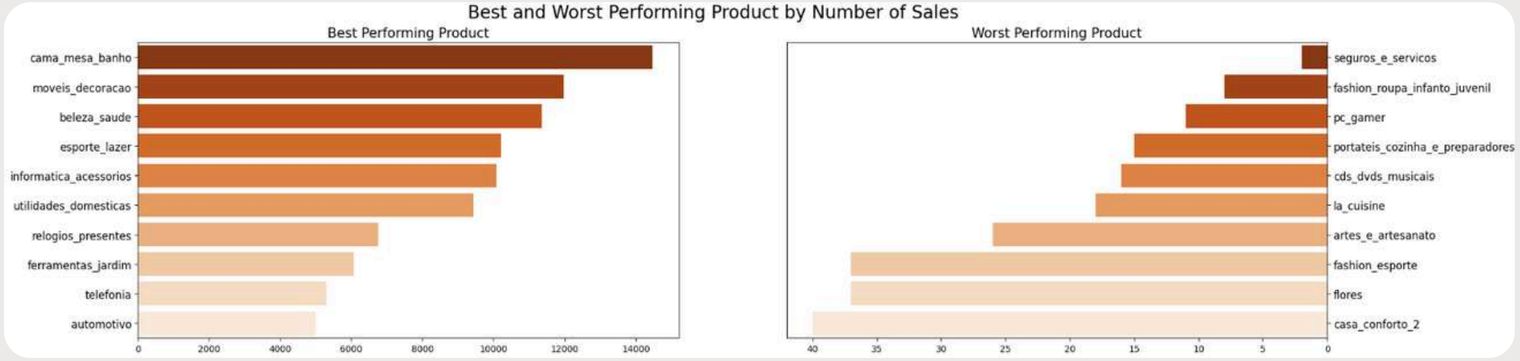
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Brazil Ecommerce Behavior

- Evaluated Brazilian e-commerce transaction trends to uncover regional consumption patterns and growth drivers.
- Identified secondary metropolitan hubs (Belo Horizonte & Curitiba) as high-potential expansion targets beyond saturated São Paulo.
- Combined product category performance with RFM segmentation to formulate data-driven market penetration and customer retention strategies.





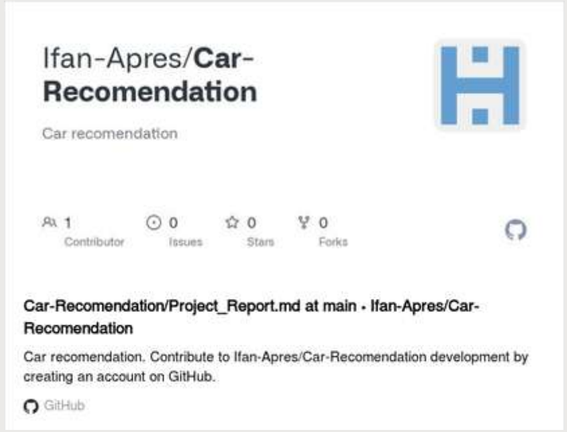


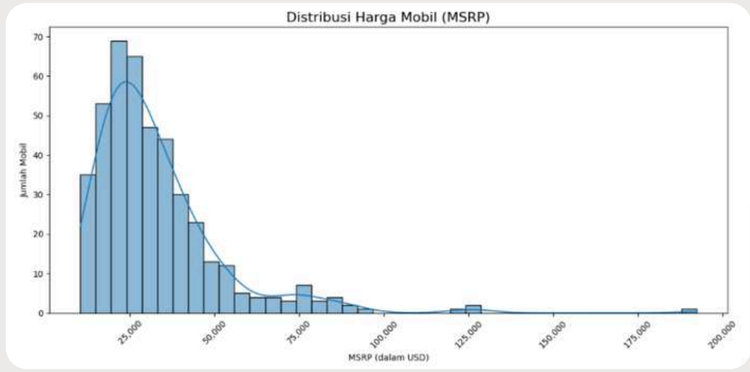
Project

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Car Recomendation

- Built a Content-Based Filtering system leveraging Cosine Similarity and K-Nearest Neighbors (KNN) to resolve consumer choice overload.
- Synthesized automotive domain knowledge with feature engineering to transform complex multidimensional vehicle specifications.
- Achieved a 71% Precision@10 benchmark, delivering objective, data-driven recommendations that guide strategic consumer financial decisions.






```
Rekomendasi untuk mobil: Viper SRT-10 convertible 2dr
--------------------------------------------------
Brand                                      Dodge
Model                Viper SRT-10 convertible 2dr
VehicleClass                             Sports
Region                                      USA
DriveTrain                                 Rear
MSRP                                    81795.0
DealerCost                              74451.0
EngineSize                                  8.3
Cylinders                                  10.0
HorsePower                                  500
MPG_City                                     12
MPG_Highway                                  20
Weight                                     3410
Wheelbase                                    99
Length                                      176
Engine_Type                              Piston
Document                 Sports USA Rear Piston
```

```
asukkan nama model mobil: Viper SRT-10 convertible 2dr
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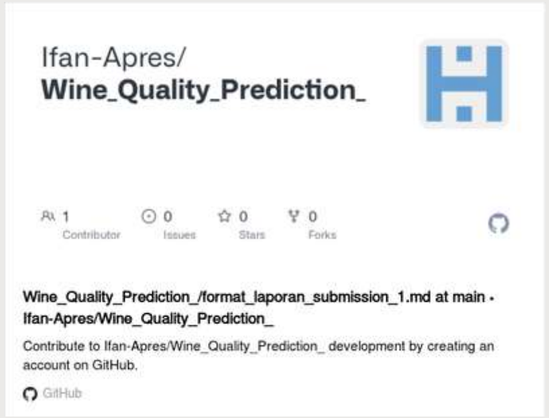
	Brand	Model	VehicleClass	MSRP	HorsePower	Engine_Type
83	Chevrolet	Corvette convertible 2dr	Sports	51535.0	350	Piston
82	Chevrolet	Corvette 2dr	Sports	44535.0	350	Piston
264	Mercedes-Benz	CLK500 coupe 2dr (convertible)	Sedan	52800.0	302	Piston
61	Cadillac	XLR convertible 2dr	Sports	76200.0	320	Piston
227	Lexus	SC 430 convertible 2dr	Sports	63200.0	300	Piston

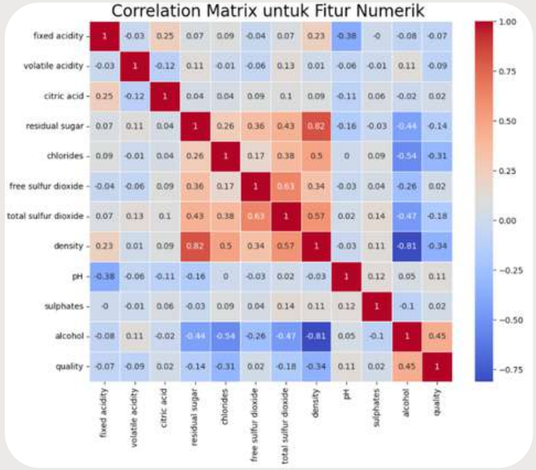
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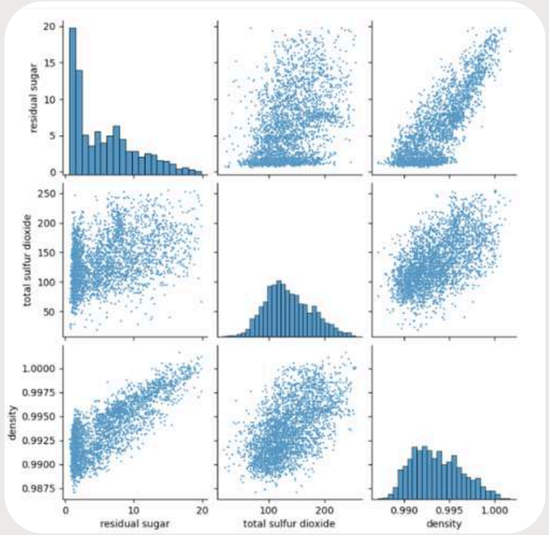
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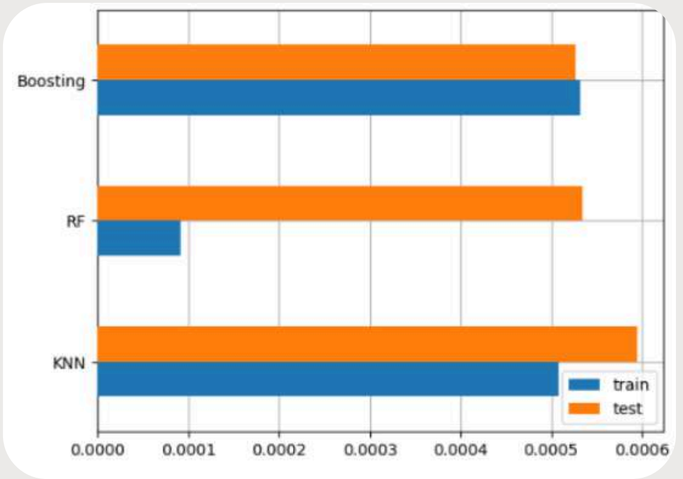
Wine Quality Prediction

- Developed an AdaBoost regression model to automate industrial quality control and mitigate manufacturing risks.
- Evaluated complex physicochemical properties using rigorous statistical feature analysis.
- Achieved an ultra-low MSE of 0.000526, delivering data-driven insights to minimize product failure losses.









Let’s connect

KHUSNIL ARIFANDI PURNAMA
MACHINE LEARNING & DATA ANALYST

Email

khusnilarifandipurnamaaa@gmail.com

Github

github.com/Ifan-Apres

Linkedin

khusnil-arifandi-purnama



The End

Khusnil Arifandi Purnama

Machine Learning & Data Analyst